Examiner only

8. *(a)* Use the information in **Figure 3** to answer the following questions.

(i) A man drinks six 175 ml glasses of wine, three pints of lager and two 25 ml glasses of spirits. Calculate the total number of units of alcohol consumed. [1]

Number of units = ...

(ii) He stopped drinking at 11 pm. He is planning to drive at 1 pm the following day. Explain whether he can be **certain** that all units of alcohol will have been processed by this time so it will be safe for him to drive. Show your calculations. [2]

....

....

(b) Use the information in **Figures 4** and **5** to answer the following questions.

(i) Explain why it is difficult to accurately compare the nutrition facts for the three cereals shown. [2]

....

....

....

(ii) Explain which cereal you would recommend for a 7-year-old child. [2]

....

....

....

(c) Calculate the daily personal energy requirement (PER) for a 70 kg athlete who trains for 4 hours a day. [3]

PER = ... units

Examiner only

(d) Use the information in **Figures 6**, **7** and **8** to answer the following questions.

(i) State the mass at which a person of height 1.7 m has a high risk of developing health problems. [1]

Mass = ... kg

(ii) Explain whether you agree with the statement that ‘BMI is proportional to body mass’. [2]

....

....

....

(iii) Calculate the BMI for a person of body mass 75 kg and height 1.6 m. [2]

BMI = ... kg/m^2

(e) (i) A woman suffers from maturity onset diabetes of the young (MODY). This is caused when one parent is heterozygous for the condition. Her partner is not diabetic. Complete the Punnett square, using **D** as the dominant allele, to show the possible genotypes of the children. [3]

(ii) State the chance of a child being born with MODY. [1]

Chance = ...

19

END OF PAPER